

How Data Work Transforms the Management of Primary Care

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Abstract

Primary healthcare institutions are incentivized to become more data-driven, as digital health data hold promises for governance and service delivery. However, data are influenced by social processes and established infrastructures, acquiring use value through so-called *data practice*, or *data work*. This PhD, a contribution to the emerging field of *data work studies*, investigates how healthcare managers utilize information systems and digital data for decision-making and planning. I'm interested in the backstage of healthcare institutions, where secondary data supports managerial processes. Adopting an ethnographic approach, I have conducted fieldwork in four Norwegian municipalities. Preliminary results suggest that data work has substantial implications for managers and their staff, requiring highly varied skillsets that are often unrecognized. This PhD project aims to enrich our understanding of managerial data practices in healthcare and how they can be improved, which is crucial for the quality of future healthcare services.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Collaborative and social computing; Collaborative and social computing design and evaluation methods; Ethnographic studies; • **Social and professional topics** → Professional topics; Computing and business; Computer supported cooperative work.

Keywords

Data work, Primary healthcare, Healthcare managers, Decision-making

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1 Introduction

With intensified focus on data in organizations, comes work with data. Data transcend mere repositories of raw information; they are

shaped by social processes and interpreted within existing socio-technical infrastructures and practices [1, 2]. Hence, data must be adapted continuously for and by their users and acquire their use value in context, through specific work situations, so-called *data practices* or *data work* [3]. *Data work studies* have become an emerging field in CSCW [4, 5]. A recent scoping review, by Bertelsen et al. [2024], concluded that data work indeed is pervasive in healthcare and that we need more research on the phenomenon to gain a deeper conceptual understanding of what data work entails in practice. Papers in this review show that data work serves various purposes, including management, clinical, and research, ensuring data quality and supporting secondary goals. Furthermore, they highlight tensions between these purposes, as information systems meant to reduce workload and increase efficiency often create more data work. For managers and policymakers, understanding these dynamics and addressing conflicts is crucial for realizing the promised values of *data-driven* healthcare [6]. Data work in healthcare means everything from registration and organization of data through daily patient work, to data analysis and strategic decisions of administrative or political nature [7]. This PhD focuses on the latter processes – data work at the backstage of healthcare services and on the use of secondary data. Such processes, known as *datafication* [8], where qualitative aspects of healthcare are converted to quantitative data for analysis, are inevitably related to power dynamics and ethics [9].

Much of the research to date has focused on the clinical frontline. My findings will extend these discussions, as I'm flipping the lens to the backstage, to the managerial context. We lack empirical knowledge on the range and diversity of data practices that exist in healthcare management [10]. Addressing this gap requires practice-based, ethnographic studies of how data work plays out among managers and their staff, with regard to both barriers and possibilities. In my PhD project, I am investigating how managers in public primary healthcare (municipalities) are utilizing information systems and digital data as foundations for decision-making regarding the services offered to the citizens. Data are seemingly stable bases for decisions, but at the same time, they often lead to hard negotiations for managers, when data are used in specific situations, what Winthereik [2024] calls *ontological trouble*. Managers in the context of this PhD project are responsible for ensuring the quality and efficiency of municipal healthcare services. They are at different management levels and oversee, sometimes multiple, assigned areas or departments of the services, managing personnel and resources, implementing changes, and coordinating with other staff and stakeholders.

The PhD is guided by the following research questions:

RQ1 - What data practices occur among municipal healthcare managers and their staff?

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RQ2 - How do municipal healthcare managers at different levels of the services work to improve data practices in their own activities, and for what purposes?

Data are strategically viewed as a key driver for public sector improvements, both as a strategic resource within public organizations [11] and as an export commodity [12]. This has been referred to as the *Nordic data imaginary* [13]. Trust in data is widespread, and some reasons for this trust can be the general high level of digitalization, alignment among public authorities, and well-connected data infrastructures [14]. Furthermore, the escalating demands on healthcare services due to an aging population, have elevated digital health data to the status of a potential gold mine in Norway, promising future benefits for governance, service delivery, research, and business enterprises [15].

The general notion seems to be that using digital data, for e.g. governance in healthcare settings, is very straightforward and easy. Davenport and Prusak [1998] claimed that a large enough data set will enable actions to objectively correct organizational decisions. This widespread optimism around data has yet to be fully realized. Schmidt and Bannon [1992] however, argued that producing and using the same set of information, requires work and effort to agree on its meaning. Human work is central to descriptions of computing, highlighting the inherent uncertainty in information sharing, as interpretation is necessary for collective use. This is in line with Berg and Goorman's [1999] *Law of medical information*, where they state that "The further information has to be able to circulate (. . .) the more work is required to disentangle the information from the context of its production." The farther data travel from their origin, the more work is required to make them meaningful. This all makes data work a complex object of research. The ones expected to do the data work [3] – to transform data into meaningful insights and actions as part of a comprehensive digital transformation – are the healthcare managers [19]. This constitutes a significant organizational challenge for computer-supported cooperative work, and there is reason to ask if the hype around data-driven management has made us underestimate the challenges. There are studies showing that data can support managers' decision-making processes and increase the continuity of healthcare services [20]. However, in many professional fields, they are already talking about data abundance – the data volume is too big to grasp [21]. This can make the appropriate use of data difficult for decision-makers. Additionally, limited knowledge among managers of how to make good use of data, presents a challenge [22].

2 Methodological Approach

I wanted to study how the use of information systems, and especially data practices, play out in naturalistic settings and therefore chose an inductive, ethnographic approach [23, 24]. Data collection takes place at the physical workplaces of the participants across multiple Norwegian municipalities, in line with the long-standing tradition of ethnographic studies in CSCW [25]. Over an eight-month period, I engaged in participant observation, semi-structured and informal interviews, and document analysis (e.g. of governance documents, case files, procedures for data use, internal and external reports). This enables me to compare and triangulate what the participants say they do in interviews, with what they actually do

when observed, and perhaps also to some extent what they are supposed to do through written instructions or regulations. Inspired by Suchman [1987], I am looking at such instructions as plans that must be adapted to concrete work situations.

Data was collected from different departments in four municipalities in Norway, differing in population sizes and in terms of digitalization needs and readiness. I recruited participants with titles of managers (e.g. head of municipal health services), middle managers (e.g. head of unit, department manager), and members of their staff supporting the management with data analysis for administration or strategic matters of the continuous planning and development of healthcare services (e.g. system coordinator, digitalization advisor). In accordance with ethnographic principles, the number of participants and sites within each municipality was decided on a case-by-case basis [27]. I recorded and transcribed the interviews, while I documented the observations through field notes and by taking photos at the sites and of situations of interest. The analysis of the data material is thematic and conducted with the help of Bernard and Ryan's [2016, 2003] techniques to identify themes and categorization.

3 Work In Progress

To date, I have completed the fieldwork, having spent time onsite different municipal healthcare departments, observing and talking to managers at different levels in their daily work practices. The analysis of the data material collected is ongoing, where I am pursuing an open-ended approach and *empirically close coding* [30], developing categories iteratively. In his book, *Data Paradoxes: The Politics of Intensified Data Sourcing in Contemporary Healthcare*, Hoeyer [2023] discusses paradoxes, or opposing stories, that stems from data work in healthcare, like *data mean less work – and data create more work*. Humans have an urge to reduce complexity, be it through the classic of closing a case with statistical argument or simplifying a narrative to fit a preconceived notion. Data work in healthcare involves endless stakeholders and purposes, which inevitably create opposing stories. These paradoxes are according to Hoeyer, the most important implications of intensified data sourcing in healthcare, and they encourage us to consider multiple perspectives simultaneously instead of choosing just one. Inspired by these thoughts, I plan to build each article in this PhD around a data work phenomenon I have uncovered, or will uncover, in my analysis, potentially originating from such paradoxes.

Since my analysis is ongoing, this paper presents preliminary insights and will not delve into any paradoxes specifically. The first finding standing out, is the skillset related to data that are required of the middle managers. My interlocutors seem to be the group that needs to handle the most information systems and data in their everyday work. Additionally, they collaborate and communicate with a range of other actors in the organizations. To be able to do all the tasks required of them, they need a most varied skillset. This includes collaborative and communicative skills but also a profound understanding of data analysis, the economics of their business, and administrative principles. As one of my informants explained, that to be able to manage this job, you are lucky if you are naturally curious about numbers and statistics, and how it all works together in the municipal ecosystem. They describe

a situation where demands from higher levels for reporting on budgets and performance seems to have increased proportionally with the numerous information systems that municipalities have invested in. During the digitalization race, nobody considered the extra workload and tasks that now come to surface. This finding is in line with recent studies that calls attention to the often-invisible extra work of handling and interpreting data [32]. Data must be generated, shaped, and presented – which requires work [3], with new jobs arising as a consequence [33].

Another finding revolves around the *extent* of data interpretation that healthcare managers must perform. One matter is the frequent data work that comes with internal reports, another is new demands for reporting on national levels. For instance, municipalities in Norway are measured and compared to each other on selected parameters, like costs in specific health posts. This creates challenges for municipalities that are seemingly very comparable to each other, but end up diverging in reported numbers, prompting questions about differences in their organizations. Data from my fieldwork suggests that such *troubles* result in new types of intra-municipal collaboration. By sitting down and discussing what really lies behind the input data and how they are interpreted on both sides, differences may also turn out not to be that significant after all. This shows yet again, that using data requires human effort to agree on its *meaning* [17]. Following Winthereik [2024], data are *relational* and lead to intense social negotiation.

My preliminary findings imply that data work in healthcare poses significant challenges for managers and their staff, as it requires additional but highly necessary work for becoming more data-driven. With many of these tasks not accounted for in the first place, this calls for an imminent change in recognition of work and collaboration around computing.

4 Expected Contributions

Through the concluding analysis, I will hopefully be able to develop some concepts that can contribute to the field. Hoeyer divides work associated with data-intensive healthcare into four main types [31]. They are *data production*, *data analysis*, *data instruction*, and *data use*, which all partly overlap. The distinction between them, however, deals with how the worker is positioned in relation to the purpose of the data work. This interpretation acknowledges that data work is carried out by a wide range of people within the healthcare system, holding highly different job titles and professional backgrounds. By building further on Hoeyer's types relevant for my context of the managerial backstage, I will zoom in on the details of how data instruction plays out in practice and understand the potential for improvement. Data instruction, according to Hoeyer [2023], is the type of work involved in governing others with, or being governed by, data. The PhD will, through this contribution, strengthen the emergent field of data work in healthcare, especially in the context of Nordic primary healthcare. Hopefully, my findings also hold relevance for developments in other, comparable health systems. More generally, it will help fill an empirical gap about data practices among healthcare administrators and managers. Furthermore, this research will hopefully contribute to a deeper understanding of governance data in public administration, in a time with increasing resource scarcity and high expectations

towards digitalization of the services. It can also highlight the invisible, but often overlooked skills required to get ahead in the digitalization race.

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